JAMDA 25 (2024) 105178

JAMDA

journal homepage: www.jamda.com

Controversies in Care

Fact Check: Assessing the Response of ChatGPT to Alzheimer's Disease Myths

Sean S. Huang MD [a,b,*], Qingyuan Song ME [c], Kimberly J. Beiting MD [a], Maria C. Duggan MD [a,d], Kristin Hines MD [a], Harvey Murff MD [a], Vania Leung MD [e], James Powers MD [a,d], T.S. Harvey PhD [f], Bradley Malin PhD [b,c,g], Zhijun Yin PhD [b,c]

[a] Division of Geriatrics, Department of Medicine, Vanderbilt University Medical Center, Nashville, TN, USA
[b] Department of Biomedical Informatics, Vanderbilt University Medical Center, Nashville, TN, USA
[c] Department of Computer Science, Vanderbilt University, Nashville, TN, USA
[d] Geriatric Research Education and Clinical Center (GRECC), Department of Veterans Affairs, Tennessee Valley Healthcare System, Nashville, TN, USA
[e] Department of Academic Internal Medicine and Geriatrics, University of Illinois at Chicago, Chicago, IL, USA
[f] Department of Anthropology, Vanderbilt University, Nashville, TN, USA
[g] Department of Biostatistics, Vanderbilt University Medical Center, Nashville, TN, USA

Keywords:
Alzheimer's disease
ChatGPT
large language models
misinformation

A B S T R A C T

Introduction: There are many myths regarding Alzheimer's disease (AD) that have been circulated on the internet, each exhibiting varying degrees of accuracy, inaccuracy, and misinformation. Large language models, such as ChatGPT, may be a valuable tool to help assess these myths for veracity and inaccuracy; however, they can induce misinformation as well.
Objective: This study assesses ChatGPT's ability to identify and address AD myths with reliable information.
Methods: We conducted a cross-sectional study of attending geriatric medicine clinicians' evaluation of ChatGPT (GPT 4.0) responses to 16 selected AD myths. We prompted ChatGPT to express its opinion on each myth and implemented a survey using REDCap to determine the degree to which clinicians agreed with the accuracy of each of ChatGPT's explanations. We also collected their explanations of any disagreements with ChatGPT's responses. We used a 5-category Likert-type scale with a score ranging from −2 to 2 to quantify clinicians' agreement in each aspect of the evaluation.
Results: The clinicians (n = 10) were generally satisfied with ChatGPT's explanations. Among the 16 myths, the clinicians were generally satisfied with these explanations, with [mean (SD) score of 1.1 (±0.3)]. Most clinicians selected "Agree" or "Strongly Agree" for each statement. Some statements obtained a small number of "Disagree" responses. There were no "Strongly Disagree" responses.
Conclusion: Most surveyed health care professionals acknowledged the potential value of ChatGPT in mitigating AD misinformation; however, the need for more refined and detailed explanations of the disease's mechanisms and treatments was highlighted.

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More than 45% of older American adults use the internet daily, with 38.9% relying on it for health information.[1] This democratization has led to increased information access and an escalation in the spread of health-related misinformation, which poses severe challenges to public health and patient decision-making.[2] Broad or lay public awareness of a health issue or a disease, including Alzheimer's disease (AD), often coexists with certain misconceptions.[3] In this respect, widely accessed AD-related statements have been posted online, demonstrating varying degrees of accuracy, inaccuracy, and misinformation. Although aware of the disease, many patients and caregivers lack specific knowledge regarding its

S.S.H. and Q.S. contributed equally to the work.

The research reported in this paper was supported by the National Institutes of Health (grant award number U54HG012510). K.J.B. is supported by the Health Resources and Services Administration (HRSA) of the U.S. Department of Health and Human Services (HHS) under grant number K01HP49070. This information or content and conclusions are those of the author and should not be construed as the official position or policy of, nor should any endorsements be inferred by HRSA, HHS, or the US government.

* Address correspondence to Sean S. Huang, MD, Division of Geriatric Medicine, Vanderbilt University Medical Center, 2147 Belcourt Avenue, Suite 100, Nashville, TN 37212.
E-mail address: sean.huang@vumc.org (S.S. Huang).

https://doi.org/10.1016/j.jamda.2024.105178
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diagnosis and management.[4] Through public access to misinformation circulated and gathered from nonclinical sources, not only is AD often unfairly stigmatized, but medical professionals are underconsulted. This trend has led some clinicians who treat AD to call for action to help distinguish fact from fiction.[5]

The emergence of large language models (LLMs) has created a new opportunity for people to search for relevant medical knowledge. LLMs comprise a complex neural network with billions of parameters estimated on large quantities of data. This platform and training process enables LLMs to capture and generate complex linguistic patterns and dependencies, thus supporting an ability to convey and express human-like text based on input prompts.[6] The commonly known LLMs, such as ChatGPT (or GPT-4), developed by OpenAI, have shown remarkable capabilities in language understanding and contextually relevant text generation.[7–12] Website traffic demographics (found on similarweb.com, accessed March 7, 2024) indicate that about 58% of users are male, 42% are female, and 25 to 34 is the largest age group. Approximately 4% of users are older than 65. However, LLMs are not infallible and do not exist outside of human culture; as such, they can mirror the assumptions and prejudices inherent in their training data, leading to biases and misinformation in their responses and raising ethical and safety concerns.[13] Thus, to leverage LLMs to combat AD misinformation, examining their performance through verification with domain or subject matter experts is necessary. ChatGPT has been shown to recognize myths in cancer,[14] but no study has shown its benefits in AD. Therefore, this study assesses ChatGPT's ability to address AD myths accurately and provide reliable information.

Methods

We conducted a cross-sectional study of clinicians' evaluation of ChatGPT's opinion of AD myths. Myths were selected from a general literature review before being independently validated as myths by 10 attending physicians who primarily practice geriatrics and completed a geriatrics fellowship. These myths were selected to encompass a wide range of topics. Myths that did not have a 10 out of 10 consensus were eliminated from the analysis. Ultimately, 16 myths were selected. These myths are listed in Table 1. In this investigation, "myths" are understood as ideas that have become accepted and otherwise understood by communities according to dominant ideas that may be based on varying degrees of accuracy, inaccuracy, and misinformation.

The study occurred between April 2023 and February 2024 and was not deemed human subjects research by the first author's (Sean S. Huang) medical center's institutional review board. Consent was obtained from all participants.

Table 1
The 16 AD Myths Used in This Study

Index	Myth
1	Symptoms of memory loss must be AD
2	All types of dementia are AD
3	All AD patients are disoriented, look confused, and act outside social norms
4	AD is a shameful diagnosis
5	An at-home genetic test can tell me if I have (or will have) AD
6	AD patients are close to death
7	AD patients cannot get depressed
8	I will develop AD if my parent has it
9	Only older people get AD
10	Supplements found online can prevent or cure AD
11	Nonsteroidal anti-inflammatory drugs can prevent AD
12	Ginkgo biloba can prevent AD
13	Hormone therapy prevents and treats AD
14	Vaccines increase the risk of AD
15	Aspartame increases the risk of AD
16	Statins increase the risk of AD

We prompted ChatGPT to generate a response to each myth. We then developed and administered a survey to determine the degree to which clinicians agreed with the accuracy of each original explanation. The survey begins with a "To ChatGPT:" component to inquire about ChatGPT's opinion on the myth. Next, the survey asks the clinician if they are satisfied with ChatGPT's opinion. The clinician could express their disagreements with the ChatGPT's opinion through a binary "yes/no" question, complemented by an optional text field to discuss the disagreements. There is also a second "To ChatGPT:" component, which contains ChatGPT's rewritten explanation and 2 questions about information retention and readability for clinicians. ChatGPT responses to all 16 myths can be found in the Supplementary Material section.

In the survey, these questions use a 5-point Likert scale that ranged from either "Very Unsatisfied" or "Strongly Disagree" to "Very Satisfied" or "Strongly Agree." Each option was associated with an according integer score ranging from −2 to 2. As such, a positive score indicates that clinicians favor ChatGPT's explanation or rewriting.

In addition, we asked clinicians before and after the survey, "Do you agree that ChatGPT can be a useful tool to provide explanations and clarifications of misinformation about Alzheimer's disease?" Each clinician was asked to review the responses for the statements in a randomized order to mitigate potential ordering effects.

The clinicians were recruited through convenience sampling and solicitation from a forum with nationwide membership. At the same time, although a diverse group of providers filled out the survey, only geriatric attending physicians who practice primarily in geriatrics and completed a geriatrics fellowship were included for analysis. Some of these clinicians are authors of this paper (6 of them, not including the first author). The surveyed clinicians practice in Arizona, Illinois, Missouri, New Jersey, New York, Tennessee, and Texas. These clinicians were asked to complete the ChatGPT assessment survey in REDCap.[15] Survey respondents received an Amazon gift card with a monetary amount proportional to the number of responses completed.

Results

Ten attending physicians who primarily practice geriatrics and completed a geriatric medicine fellowship completed the survey. Of these, 9 completed all the survey questions and provided valid answers. Only one of these clinicians had used ChatGPT occasionally, 9 clinicians had heard of it but never used it, and 1 other clinician had never heard of it.

Figure 1 illustrates the participants' evaluation of ChatGPT's opinions. Generally, among the 16 myths, the clinicians were satisfied with these explanations with [mean (SD) score of 1.1 (±0.3)]. For reference, a score of 1 indicates agreement and 2 indicates strong agreement. Most clinicians selected "Agree" or "Strongly Agree" for each statement (green in Figure 1). Some statements showed a sparse amount of "Disagree" responses. There were no "Strongly Disagree" responses (red in Figure 1).

We compared the clinicians' assessment of ChatGPT's potential to clarify and explain misinformation about AD before and after reviewing ChatGPT's responses. One attending had never heard of ChatGPT, and another had used it occasionally; both agreed it was a useful tool to explain and clarify misinformation about AD before and after the 20 surveys. The other attending physicians had heard about ChatGPT but never used it.

Two attending physicians had initial responses of "Disagree." One of them changed their response to "Neither Agree nor Disagree (Neutral)," and the other changed their response to "Agree." Three others changed their responses from "Neutral" to "Agree." Another 2 changed their responses from "Agree" to "Strongly Agree."

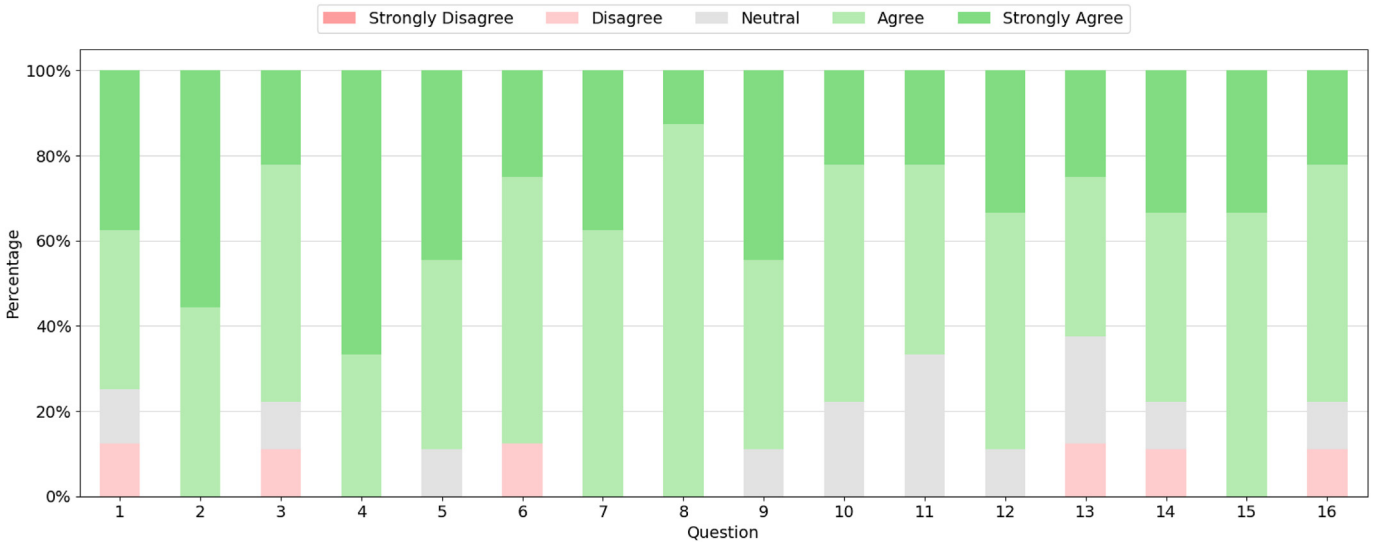


Fig. 1. Agreement and disagreement for each ChatGPT explanation for each myth among the 10 participants.

Discussion

This investigation suggests that ChatGPT can identify and explain misinformation about AD and generate audience-specific readable explanations. Still, several explanations needed more accuracy, and disagreement was present (although not prominent).

There are limitations to the study. First, the clinicians who evaluated the statements were limited to only geriatric attendings. We thought limiting scope was important for explainability and establishing the geriatrician's role in assessing the utility of LLMs. Adding non-geriatricians and non-attending providers may lead to unnecessary nuance in our analysis. These comparisons can be made in future work.

There was also a bias toward providers already interested in the topic. Clinicians were chosen based on responses to a message board forum who knew that the study was about AD and LLMs. Being able to ask less technologically inclined clinicians may make the study more impactful. The myths were chosen to encompass many subtopics rather than represent the most common myths.

Although Likert scales are widely used in surveys to measure attitudes and opinions, they come with definite limitations. There can be a central tendency bias toward extreme response categories. Simply having agree/disagree may solve this issue at the cost of less nuanced responses. The interpretations of the categories can be ambiguous and limited. Finally, Likert scales produce ordinal data that can also limit the analyses that can be performed.

To gain insight into the scenarios in which ChatGPT excelled, we examined 2 statements with the highest total Likert-type scores among our 16 myths.

Myth #4: AD Is a Shameful Diagnosis

Average Likert-type score: 1.67
ChatGPT. "As an AI language model, I don't have personal opinions or feelings. However, I can share that there is no reason to consider Alzheimer's disease or any other medical condition as a 'shameful diagnosis.' Alzheimer's disease is a neurodegenerative disorder that affects memory, thinking, and behavior. It is a medical condition that should be treated with compassion, understanding, and support.

Stigmatizing Alzheimer's or any other medical condition can have harmful effects on individuals with the disease and their families, as it may lead to isolation, delay in seeking help, and diminished quality of life. It is essential to raise awareness about the disease, promote understanding, and support research to improve treatment options and quality of life for those affected by Alzheimer's disease."

AD often carries a social stigma that produces harmful effects on the well-being of patients afflicted with the disease and their families. Under-recognition and under-treatment by providers often lead to delays in care-seeking, diagnosis, and accessing resources for treatment. Other reasons include professional biases and avoidance of discussion by providers due to stigma. The LLM prefaces its statement by noting that it does not have a personal opinion. Because ChatGPT is not a person, this statement is technically true. However, the statement can be interpreted as amounting to a claim of objectivity. That said, ChatGPT expresses the opinion that AD is not a shameful diagnosis and those affected should be treated with compassion, understanding, and support.[16] Although no diagnosis should ever be "shameful," it is essential to note that technically, what makes a diagnosis "shameful" is the social context in which the disease is experienced and not the diagnosis itself. Its anticipated response echoes recent suggestions that ChatGPT and similar models can mirror empathy equivalent to a human's.[17] Although it certainly cannot replace a clinician trained in managing AD or caregiver communities of practice, working LLMs may one day be used to establish more personalized and potentially less stigmatized care.

Myth #2: All Types of Dementia Are AD

Average Likert-type score: 1.56
ChatGPT. "No, this statement is incorrect. Although Alzheimer's disease is the most common type of dementia, accounting for 60-80% of cases, there are many other types of dementia. Dementia is an umbrella term for a group of symptoms related to cognitive decline, which affect memory, thinking, problem-solving, and other cognitive functions.

Some other common types of dementia include.

- Vascular dementia: This is the second most common type of dementia and is caused by impaired blood flow to the brain, often due to a stroke or a series of small strokes.
- Lewy body dementia: Characterized by the presence of abnormal protein deposits called Lewy bodies in the brain, this type of dementia is the third most common and has symptoms

that can resemble both Alzheimer's disease and Parkinson's disease.
- Frontotemporal dementia: This type of dementia primarily affects the frontal and temporal lobes of the brain, which are responsible for behavior, language, and personality.
- Mixed dementia: This refers to the presence of more than one type of dementia in an individual, such as a combination of Alzheimer's disease and vascular dementia.

There are also many other less common types of dementia and conditions that can cause dementia-like symptoms. It is important to receive an accurate diagnosis for the specific type of dementia, as treatment and management strategies can vary depending on the underlying cause."

ChatGPT makes a very emphatic declaration, appropriately noting that this statement is incorrect. It notes that AD is the most common type and then demonstrates that not all types of dementia are AD by identifying other types of dementia, including vascular dementia, Lewy body disease (LBD), and frontotemporal dementia.[18] The distinctions are important as they may affect care planning and treatment decisions. For example, cholinesterase inhibitors are commonly effective in LBD but can often worsen frontotemporal dementia. ChatGPT only mentions that symptoms of LBD can resemble Parkinson's disease but fails to distinguish that dementia with Lewy bodies and Parkinson's disease dementia are 2 different diseases with different underlying etiologies, pathologies, and diagnostic criteria.[19,20]

Implications for Practice, Policy, and Research

Like many diseases, AD, given its prevalence, is prone to misinformation, a problem that is exacerbated and accelerated by online platforms. LLMs, like ChatGPT, are generally not directly responsible for creating or propagating misinformation. However, their utilization can inadvertently amplify it due to potential biases or misinterpretations in their training data. Surprisingly, ChatGPT did reasonably well in answering questions about AD, corroborated by attending geriatrics clinicians. Although ChatGPT can provide helpful information for most readers, there could be more refined and detailed explanations of the disease's mechanisms and treatments.

Disclosure

The authors declare no conflicts of interest.

Acknowledgments

Dr. Yin (sponsor) played an integral role in the design, methods, analysis, and preparation of the paper.

Supplementary Data

Supplementary data related to this article can be found online at https://doi.org/10.1016/j.jamda.2024.105178.

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